

MEASURING THE FOURTH FUNCTION

An Empirical Companion: The Scholarly, Judicial, and Regulatory Uptake of Federal-Agency Research

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DRAFT — FOR DISCUSSION

June 2026

ABSTRACT

This is the empirical companion to *The Fourth Function: Research, Data, and Analysis as a Power of the Administrative State*. Where the Note argues that federal agencies exercise a distinct, under-governed research function, this companion measures one of its premises—that agency research has large and unevenly distributed downstream effects—using three free, public, reproducible data sources: OpenAlex for scholarly citation, CourtListener for the judicial record, and the Federal Register API for the rulemaking record. Three findings emerge. First, the Federal Reserve’s scholarly footprint is enormous and exceeds that of the statistical and economic-analysis agencies combined; by citation count the Board of Governors ranks among the largest economics-research institutions in the world. Second, that scholarly dominance inverts in the courts: the statistical agencies’ products—the Consumer Price Index, Bureau of Labor Statistics data—appear in thousands of judicial opinions, while the Federal Reserve’s flagship research products appear in almost none, exactly as the Note’s “accountability paradox” predicts. Third, analytical language is a stable and pervasive feature of the rulemaking record across three decades rather than a recent development. The companion then specifies, and previews, a lead-lag design to test the hypothesis that much agency research is reactive to policy attention rather than antecedent to it, and closes with a candid account of the measurement’s limits. All data and code accompany this draft.

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Introduction

The companion Note to this paper makes a doctrinal argument: that the production of research, data, and analysis is a fourth function of the administrative state, distinct from rulemaking, adjudication, and enforcement, and largely ungoverned by administrative law.¹ That argument rests on an empirical premise—that agency research has impact—and on a sharper subclaim, the *accountability paradox*: that the legal control exerted over a body of agency research tracks its legal operativeness rather than its real-world influence, so that the most consequential research is often the least constrained. This companion tests what can be tested of those claims with public data.

The exercise is deliberately modest in its tools and transparent in its limits. It does not attempt a causal estimate of the effect of agency research on policy outcomes. It measures *uptake*: the degree to which the research of three clusters of agencies—the Federal Reserve, the federal statistical system (the Census Bureau and the Bureau of Labor Statistics), and the economic-analysis agencies (the Securities and Exchange Commission and the Federal Trade Commission)—propagates into three corpora that can be counted: the scholarly literature, the body of judicial opinions, and the rulemaking record. Uptake is a proxy for influence, and an imperfect one; but the disparities it reveals are sufficiently large, and sufficiently patterned, to bear the weight the Note’s argument places upon them.

Part I describes the data and method. Part II reports the results across the three corpora. Part III specifies the lead-lag design for the reactive-literature hypothesis and previews its logic.

¹*The Fourth Function: Research, Data, and Analysis as a Power of the Administrative State* (2026) [hereinafter Note]. This companion is designed to be read after the Note, whose Part VI summarizes the headline results reported here in full.

Part IV sets out the threats to inference. A short conclusion connects the findings back to the Note’s doctrinal claims.

I. Data and Method

A. *Three Corpora, Three Free Sources*

The design measures agency-research uptake in three corpora, each through a free public interface that imposes no subscription cost and requires no proprietary access.

Scholarly literature — OpenAlex. OpenAlex is an open bibliographic database that catalogs scholarly works, their authors and institutional affiliations, and the citations among them. For each agency research arm, the companion retrieves the institution’s catalogued output (works), the cumulative citations to that output, and standard bibliometric summary statistics, including the h-index.²

Judicial record — CourtListener. CourtListener, maintained by the Free Law Project, is a comprehensive open database of American judicial opinions. For each distinctive research product, the companion counts the opinions whose text contains the product’s name, in total and by decade.³

Rulemaking record — the Federal Register. The Federal Register API exposes the full text of rules, proposed rules, and notices. For each term, the companion counts the documents whose text contains it, in total and by year.⁴

²OpenAlex, api.openalex.org. Institutions were resolved by name search and confirmed by country and output volume; counts were retrieved through the institutions and works endpoints under OpenAlex’s polite-pool conventions (a contact address is supplied with each request). Retrieval date: June 2026.

³CourtListener, courtlistener.com, REST API v4. Counts are exact-phrase searches over the opinion corpus; the API returns a total match count and supports date-filtered subqueries by filing date. CourtListener’s coverage is broad but not exhaustive, and varies by jurisdiction and era. Retrieval date: June 2026.

⁴Federal Register API, federalregister.gov/developers/api/v1. The API caps reported totals at 10,000; for the most common terms the annual series, which is uncapped at the values observed here, is the reliable measure. Retrieval date: June 2026.

B. Measures and Their Limits

The unit of analysis throughout is the document, and the measure is a count. This simplicity is a virtue for reproducibility and a limitation for precision, and that limitation should be acknowledged at the outset. A count of mentions is not a measure of influence: it cannot distinguish a citation that carries an argument from one that decorates it, and it misses influence that travels without naming its source. Three specific artifacts recur and are flagged wherever they bear on a result.

First, *affiliation attribution*. OpenAlex assigns a work to an institution whenever any contributing author lists that affiliation. This over-counts: the single most-cited work attributed to the Federal Reserve Board in the raw data is a scientific-computing software paper, not economics research, included because a contributor listed a Federal Reserve affiliation.⁵ Second, *the Federal Register cap*, which fixes reported totals at ten thousand and makes the annual series the trustworthy measure for common terms. Third, *phrase ambiguity*: a term like “economic analysis” counts all usages, not only those referring to a particular agency’s research. None of these artifacts can convert the order-of-magnitude differences reported below into artifacts of measurement, but each counsels against reading small differences as real.

C. Reproducibility

The entire pipeline is scripted and cached. A collection script retrieves the data from the three APIs and stores the raw responses as JSON; a plotting script renders the figures from that cache; the counts reported in the tables are read from the same cache. Re-running the collection

⁵*SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python* (2020), which alone accounts for roughly five percent of the Board’s catalogued citations. The aggregate counts remain informative as orders of magnitude; individual point estimates should not be over-read.

reproduces the dataset (subject to the corpora having grown since retrieval), and re-running the plotting reproduces every figure exactly. No step depends on a paid service or a credential.⁶

II. Results

A. Scholarly Footprint

By every bibliometric measure, the Federal Reserve is the dominant research producer among the agencies studied. The Board of Governors alone has a catalogued output of more than fourteen thousand works and has accumulated roughly 759,000 citations—more than the Census Bureau, the Bureau of Labor Statistics, the SEC, and the FTC combined—with an h-index in the hundreds that places it among the most-cited research institutions of any kind. Adding the New York and St. Louis Reserve Banks brings the Federal Reserve system’s catalogued citations past one million. Table 1 reports the figures; Figure 1 displays the citation comparison.

Table 1. Scholarly footprint of agency research arms (OpenAlex, retrieved June 2026).

Institution (cluster)	Works	Citations	h-index
Federal Reserve Board (Fed)	14,345	759,106	333
FRB New York (Fed)	4,587	295,443	239
U.S. Census Bureau (statistical)	4,722	172,756	182
FRB St. Louis (Fed)	4,202	113,556	138
Bureau of Labor Statistics (statistical)	2,706	91,650	107
SEC (economic-analysis)	1,320	41,478	91
FTC (economic-analysis)	1,610	36,396	90

⁶The accompanying repository contains the data-collection script, a gentle backoff-aware re-fetch for rate-limited endpoints, the figure-plotting script, and the cached JSON responses. The throttling and contact-address conventions are set to honor each service’s published use policy.

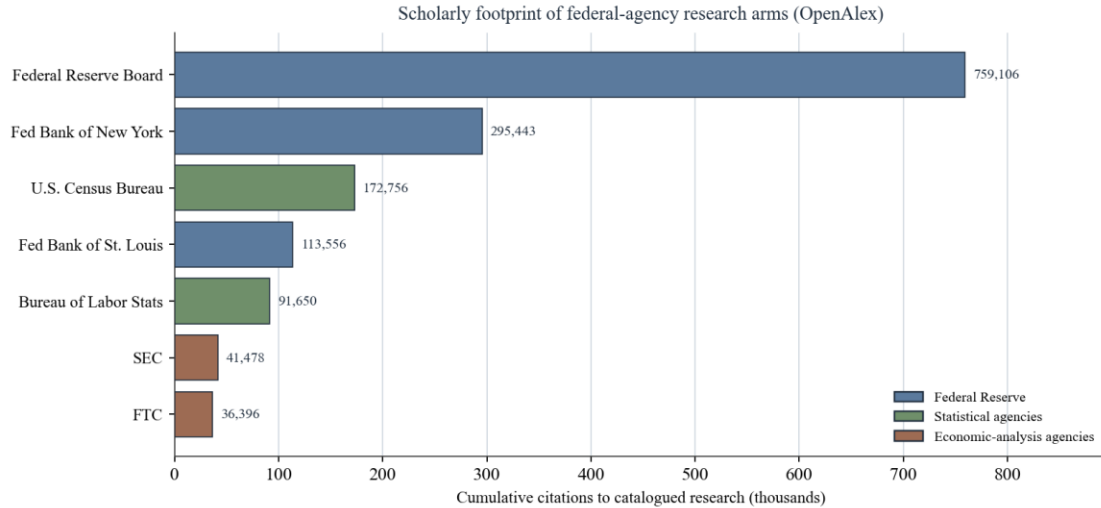


Figure 1. Cumulative scholarly citations to each agency’s catalogued research output (OpenAlex). Colors denote the three clusters. Institution-level counts are subject to the affiliation-attribution caveat discussed in Part I.B.

The result confirms a claim the Note makes qualitatively: the Federal Reserve is not merely a central bank that does research but, by the metrics of the scholarly economy, one of the largest economics-research institutions in the world. The disparity is not subtle—the Board’s citation count is an order of magnitude larger than the FTC’s—and it is robust to the affiliation caveat, which at five percent cannot explain a gap of that size.

B. Judicial Uptake

In the courts the ranking inverts. The research products that appear most often in judicial opinions are the legally operative statistics and the analysis that conditions rulemaking, not the Federal Reserve’s celebrated research. The Consumer Price Index appears in some 3,400 opinions and Bureau of Labor Statistics data in roughly 1,900; “cost-benefit analysis” appears in some 2,700. The Federal Reserve’s flagship research products are by comparison invisible: its triennial Survey of Consumer Finances appears in four opinions, the Beige Book in none, and the Senior Loan Officer Opinion Survey in none. Table 2 reports the counts; Figure 2 illustrates the contrast.

Table 2. Judicial opinions mentioning each research product (CourtListener, retrieved June 2026). The 2020s column is partial (through mid-2026).

Product (cluster)	Total	1990s	2000s	2010s	2020s
Consumer Price Index (stat.)	3,397	701	718	812	202
cost-benefit analysis (econ.)	2,725	467	617	775	321
Bureau of Labor Statistics (stat.)	1,932	190	298	476	181
American Community Survey (stat.)	142	0	13	95	34
FTC Bureau of Economics (econ.)	108	10	30	20	2
Survey of Consumer Finances (Fed)	4	0	2	0	1
SEC Div. of Econ. & Risk Analysis (econ.)	4	0	0	0	4
Beige Book (Fed)	0	0	0	0	0
Senior Loan Officer Survey (Fed)	0	0	0	0	0

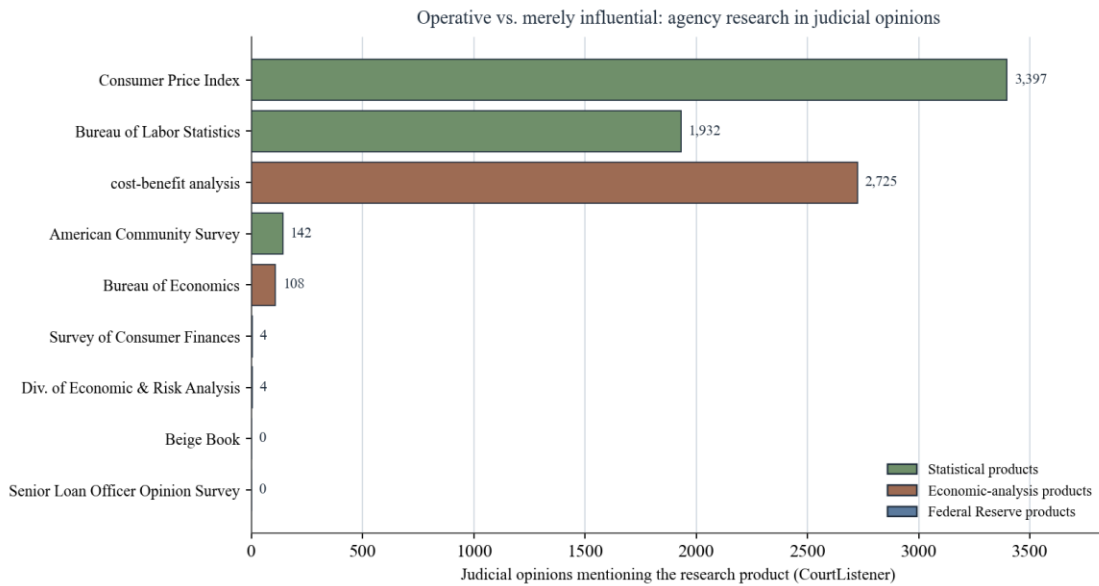


Figure 2. Judicial opinions mentioning each research product (CourtListener). Statistical and economic-analysis products that bear on legal consequences appear thousands of times; the Federal Reserve's flagship research products are nearly absent.

This is the accountability paradox observed directly. Research that is legally operative—the statistics wired into benefits, taxes, and apportionment, and the analysis welded to reviewable rules—appears where legal consequences are decided, because it helps decide them. Research that is merely influential—however vast its scholarly footprint—does not surface in litigation, because it operates through channels the courts do not police. The Federal Reserve has the largest research enterprise and the smallest judicial footprint of the three clusters, the clearest illustration of the divergence between influence and legal operativeness on which the Note's argument turns.

The decade breakdown adds a temporal dimension, shown in Figure 3. The judicial presence of the established statistical products is durable across forty years; the American Community Survey rises from literal nonexistence before its mid-2000s launch to a regular presence in the courts within a decade—a clean natural illustration of a new agency research product passing into regular judicial use.⁷

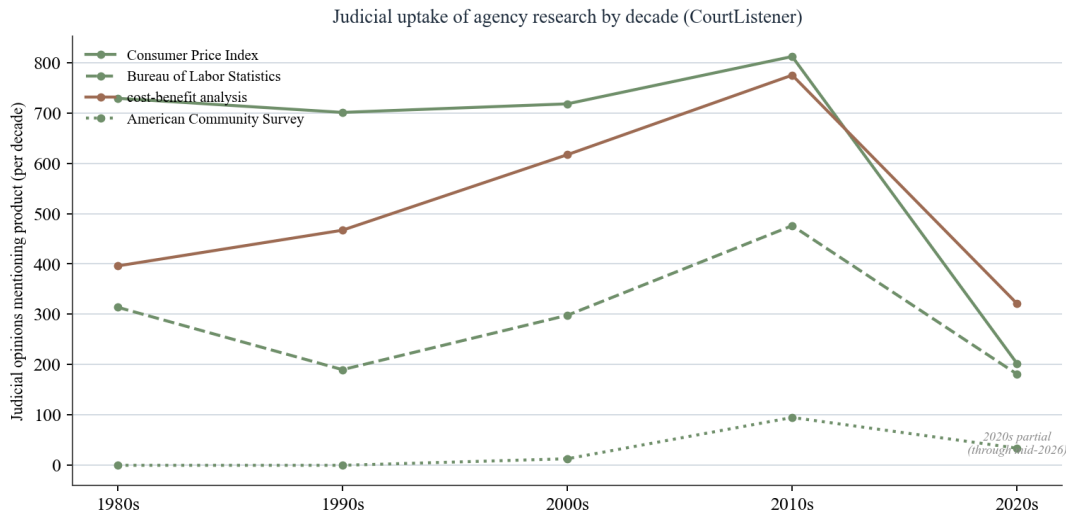


Figure 3. Judicial uptake of agency research products by decade (CourtListener). The 2020s point is partial (through mid-2026). The American Community Survey enters the courts only after its mid-2000s launch.

C. Regulatory Uptake

In the rulemaking record, analytical language is pervasive and, strikingly, stable. The phrase “economic analysis” appears in thousands of Federal Register documents in a typical year, fluctuating roughly between 2,500 and 3,600 annually from 1995 through 2025 without a clear secular trend; “cost-benefit analysis” appears in a smaller but similarly steady band. Specific research products track the salience of their underlying programs: the Consumer Price Index and the American Community Survey appear in hundreds of documents per year, the Survey of Consumer Finances in a handful. Table 3 reports totals and Figure 4 the annual series.

⁷The 2020s figures are partial, covering filings through mid-2026, and so understate the full-decade totals; the apparent decline in the last point of each series in Figure 3 is an artifact of the truncated window, not a real reversal.

Table 3. Federal Register documents mentioning each term (retrieved June 2026). Totals marked “10,000+” are at the API’s reporting cap; for those terms the annual series is the reliable measure.

Term (cluster)	Total (capped at 10,000)	Typical annual count
economic analysis (econ.)	10,000+	~2,500–3,600 / year
Consumer Price Index (stat.)	9,881	~200–500 / year
cost-benefit analysis (econ.)	8,663	~150–450 / year
American Community Survey (stat.)	10,000+	~300–750 / year
Survey of Consumer Finances (Fed)	773	~10–50 / year

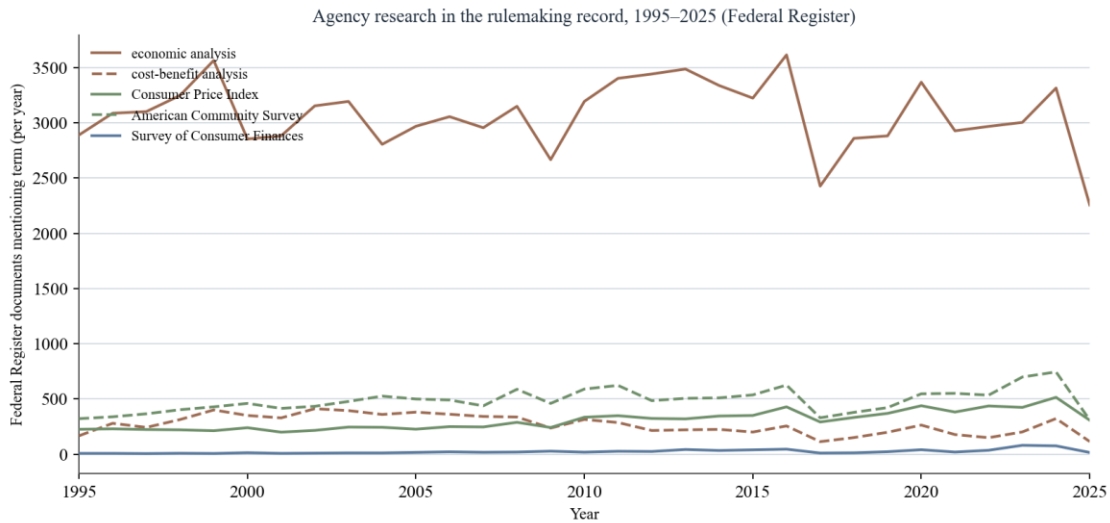


Figure 4. Federal Register documents mentioning selected research terms, by year, 1995–2025. “Economic analysis” is a constant fixture of the rulemaking record; product-specific references are smaller and program-driven.

The stability is the notable result. A common narrative dates the “cost-benefit state” to a series of developments in the 2000s and 2010s—the judicial cost-benefit cases, successive executive orders on regulatory review. The Federal Register data suggest the analytical turn is older and more entrenched: the rulemaking record’s appetite for analysis has been roughly constant for three decades. Whatever the courts and the executive orders did, they operated on a regulatory culture in which analysis was already ubiquitous.⁸

⁸This is consistent with, though it does not prove, the Note’s suggestion that the economic-analysis model is becoming the template for administrative law generally rather than a recent innovation. *See* Note, Part IV.C. The flat trend also cautions against attributing the prevalence of analysis to any single doctrinal or executive intervention.

III. The Reactive-Literature Hypothesis

The descriptive results establish uptake. They do not establish direction: whether agency research leads the policy and public attention it addresses or follows it. The conventional account assumes leadership—the agency produces knowledge, and policy then acts on it. The reactive-literature hypothesis is the contrary conjecture: that a substantial share of agency research is produced and cited in response to a controversy or a pending decision, rather than in advance of it. The distinction matters for the Note’s normative argument, because the case for insulating agency research from political control is strongest when the research is genuinely antecedent and weakest when it is mobilized to support a position already taken.

The design to test the hypothesis is a cross-correlation, or lead-lag, analysis. For a given research topic, one constructs three time series: the production of agency research on the topic (proxied by the OpenAlex publication-year counts for the relevant works), the policy attention to it (proxied by Federal Register mentions by year), and the judicial attention to it (proxied by CourtListener filings by year). One then estimates the cross-correlation of the research series against each attention series at a range of leads and lags. If research *leads*, the correlation peaks when the research series is shifted earlier in time; if research *lags*—the reactive case—the correlation peaks when the research series is shifted later.

The companion implements this design at the topic level rather than reporting it here, for a reason the next Part develops: the timing artifacts are severe enough that the estimates should be presented as exploratory, with placebo windows and alternative lag specifications, rather than as a headline finding. What the descriptive data already suggest, and what the lead-lag design is built to test rigorously, is that the conventional account, in which research leads policy, is at best incomplete. The flat trajectory of the Federal Register analysis series, for instance, is hard to square with a story in which research drives a growing analytical demand; it is easier to square with a

story in which analysis is produced to satisfy a standing institutional expectation—reactively, on demand.⁹

IV. Threats to Inference

The descriptive findings are robust because they are gross: the gaps they report are of orders of magnitude, and no plausible measurement correction reverses them. Four opinions citing the Survey of Consumer Finances do not become four thousand under any reasonable adjustment; the Board’s scholarly dominance survives the removal of mis-attributed works many times over. For the descriptive claims, the threats catalogued below affect precision, not direction.

The causal hypothesis is far more fragile, and candor requires that the principal threats be named. *Publication lag* is the most serious: scholarship responding to a 2020 event may not appear until 2022 or 2023, which can make genuinely reactive research look contemporaneous with, or even anticipatory of, a later event. *Corpus lag* compounds it: the Federal Register and the judicial record have their own institutional delays, so the three series are not on a common clock. *Phrase ambiguity* means the research and attention series may not track the same underlying topic. And *coverage variation*—CourtListener’s uneven coverage across jurisdictions and eras, OpenAlex’s changing completeness over time—can induce spurious trends.¹⁰

The honest summary is that the public data are sufficient to establish the Note’s factual premise—that agency research has large and unevenly distributed downstream effects—and to motivate, but not yet to prove, the reactive-literature hypothesis. That asymmetry is itself a useful

⁹The reactive-literature framing connects to a companion line of inquiry into whether scholarly and official literatures lead or lag the policy developments they discuss. The present design is the first cut; a fuller treatment would incorporate topic models to define the research and attention series consistently and would exploit discrete policy shocks to sharpen identification.

¹⁰The companion’s mitigations include: restricting lead-lag estimates to topics with well-defined, unambiguous product names; using placebo event windows to gauge the false-positive rate; reporting results under multiple lag conventions; and treating the earliest years of each corpus, where coverage is thinnest, with caution. Even so, the causal estimates are offered as exploratory.

result: the descriptive case for taking the research function seriously as an object of administrative law does not depend on resolving the harder causal question, and can be made on the counts alone.

Conclusion

Three free databases, interrogated with nothing more elaborate than counts, suffice to give quantitative form to the Note's central claims. The Federal Reserve's research is the most prolific and most-cited in the federal government, by a substantial margin—and almost entirely absent from the judicial record, where the statistical agencies' legally operative numbers predominate. Analytical language pervades the rulemaking record and has done so steadily for thirty years. Together these patterns are the accountability paradox in empirical form: influence and legal control come apart, most dramatically at the most insulated agency. The harder question—whether agency research leads policy or reacts to it—remains open, and the companion's lead-lag design is a first attempt on it rather than a settled answer. But the descriptive case is already made, and it is the case the Note needs: the fourth function is real, it is large, and the law that reaches it does not track its importance.